

Eshwar Pamula

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EDUCATION

The Ohio State University - Honors Scholar

Columbus, OH

Bachelor of Science in Mechanical Engineering with Research Distinction

Graduation: December 2025

Minor: Economics

GPA: 3.37/4.0

WORK EXPERIENCE

Flux Marine | Powertrain Engineering Intern

May 2025 - August 2025

- Contributed to the design and development of the stator for the second-generation in-house electric motor prototype
- Developed a fixture with a custom quick-release clamp mechanism for stator segments, ensuring consistent motor assembly
- Designed and built a shrink-fit jig to reduce assembly process time by 30% while guaranteeing flush seating of all segments
- Designed and manufactured a coil winding fixture for the stator segments, for controlled tension to ensure consistent, tight winding of the coils without gaps or overlaps
- Ran comprehensive dynamometer testing across varying torque and RPM ranges to validate motor performance metrics
- Contributed to cross-functional product development across 3 different motor builds at a Series-A marine propulsion startup

Multi-functional Materials and Intelligent Design Lab | Undergraduate Researcher

May 2024 - Present

- Awarded a College of Engineering grant for AI-driven geometric primitive-based topology optimization
- Developed machine learning algorithms in Python to extract geometric primitives from topology-optimized structures, reducing computational solving time by identifying reusable structural patterns

Allegion Inc. | R&D Mechanical Engineering Intern

May 2024 - August 2024

- Worked on new product development initiatives for two new electro-mechanical residential lock set designs
- Designed a dual-CAM interconnect assembly using CREO and validated it through FEA simulations for BHMA standards
- Helped develop patent-pending housing for a new lock-set design set to be launched in early 2026
- Manufactured early prototypes (3D printing and CNC) of the lock system, ensuring feasibility for the product pipeline

Gear and Power Transmission Laboratory | Undergraduate Research Assistant

April 2023 - May 2024

- Conducted experimental analysis of Torlon gear properties using Symbrium machines
- Monitored and controlled oil flow, temperature, and torque conditions to assess thermal performance of gear systems

Ecosystem Energy | Mechanical Engineering Intern

May 2023 - August 2023

- Monitored and analyzed HVAC control data from heat exchangers to enhance energy efficiency and system optimization
- Designed energy control systems in AutoCAD, collaborated with three cross-functional teams, and oversaw on-site construction for a \$300K pneumatic valve project, achieving a 27% reduction in energy consumption

Engie North America | Project Engineering Intern

October 2022 - May 2023

- Conducted engineering reviews of HVAC designs using AutoCAD for Ohio State's Combined Heat and Power plant

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

Formula Buckeyes FSAE | Suspension Engineer & Business Lead

2022 - August 2024

- Helped raise over \$35K and sponsored parts from automotive OEMs and Mercedes for building the car
- Designed and manufactured uprights, mounts, and hard points using the mill and lathe for the control arms and sway bar
- Helped manufacture in-house 7075-T6 Uprights using Kia SKT25 CNC
- Contributed to the design, manufacture, and testing of a Front Anti-Roll Bar (T-Bar setup) with SolidWorks and Lathe CNC
- Played a key role in the team's success, securing a 3rd-place finish in 2023 and a 1st-place finish in 2024 at Michigan FSAE

The Ohio State University: MAE | Undergraduate Teaching Assistant

April 2023 - Present

- Taught and mentored 500+ students over five semesters in Statics, Mechanics of Materials, and Machine Elements II
- Awarded Best Undergraduate Teaching Assistant in the MAE Department (2025)

AWARDS, SKILLS, AND INTERESTS

- Awards:** Best Teaching Assistant ('25), Engineering Showcase Winner ('23), OSHA 10, NASBLA Boating License
- Programming Languages:** MATLAB, Python, Arduino IDE, HTML/CSS
- CAD & FEA Software:** SolidWorks, PTC OnShape, CREO, AutoCAD, ANSYS, ABAQUS, SolidWorks MasterCAM
- Skills & Tools:** 5 Axis CNC, Lathe, 3D Printing, Mill, GD&T
- Interests:** Film Photography (35mm), Piano, Soccer, Table Tennis